Global Gauss–Bonnet Theorem for Surfaces

An End-to-End Meta-Math Walkthrough

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Abstract

A complete authored walkthrough of global gauss–bonnet theorem for surfaces, with explicit mathematical contracts, local verification gates, and reusable exports. The report presents the target theorem, declared imports, a dependency graph, five-component node contracts, a continuous proof, a worked example, and a hypothesis-repair audit. It is an authored mathematical exposition, not a deployed-agent transcript or a formal proof-assistant certificate.

DG-04 / Differential Geometry / Report v1

I. TARGET AND IMPORT CONTRACT

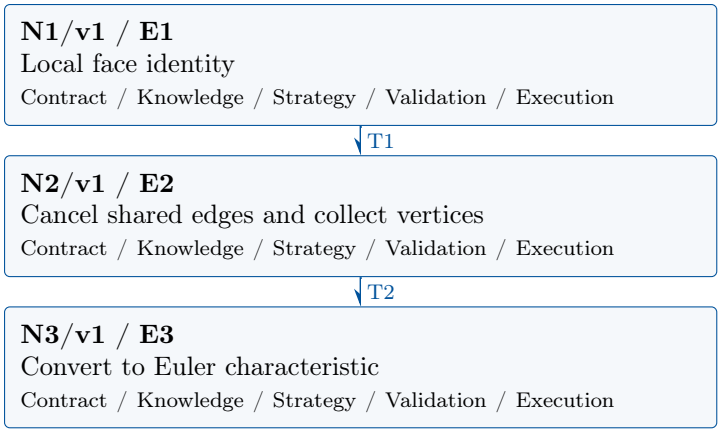
Theorem I.1. *For a closed oriented smooth Riemannian surface M, prove $\int_M K\,dA = 2\pi\chi(M)$. Here closed means compact without boundary.*

A. Foundations and permitted imports

Import local Gauss–Bonnet for an oriented disk-like region with piecewise smooth positively oriented boundary: $\int K\ dA + \int boundary\ k_g\ ds + \sum$ exterior turning $angles = 2\pi$. Also import a finite smooth triangulation of the closed oriented surface with regular edges and tangent sectors $filling 2\pi$ around every vertex. Curvature is intrinsic by DG-03. These local geometric and triangulation modules are substantial imports; the global formula is proved by assembly.

For an abstract Riemannian surface, K is defined intrinsically from its Levi-Civita curvature tensor. DG-03 explains the agreement with extrinsic Gaussian curvature on embedded surfaces; its extrinsic statement is context, not a necessary premise of this assembly proof.

II. NODE INTERACTION WORKFLOW



Each box is one mathematical agent containing all five functional components. Blue arrows carry the versioned exports listed below. The layout gives a reading order; only drawn arrows assert dependencies. Diagnostic repair is separate from the proof DAG.

A. Versioned evidence-transfer ledger

T1: N1/E1/v1 to N2. A local curvature identity for each face.

Receiving activity. N2: Cancel shared edges and collect vertices

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: All local face identities; closed oriented surface; prior exports E1/v1

T2: N2/E2/v1 to N3. The global integral equals $2\pi V - \pi F$.

Receiving activity. N3: Convert to Euler characteristic

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: Triangulation counts; prior exports E2/v1

III. AGENT ROLES AND FIVE-COMPONENT ACTIVITIES

A. N1/v1: Local face identity

Object and role. Express turning angles through interior face angles.

Inputs. A compatible finite triangulation; foundational imports stated in §1

C — Contract

Own E1/v1: A local curvature identity for each face.

K — Knowledge

Retain A compatible finite triangulation; foundational imports stated in §1; keep their assumptions and versions attached.

S — Strategy

Express turning angles through interior face angles.

V — Validation

Do not silently assume every triangulation edge is geodesic.

E — Execution

Carry out the numbered deductions and publish E1/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N1.1 For each triangular faceF with interior corner angles θ_{Fv}, the local theorem reads $\int_F KdA + \int_\partial F\ k_g ds + \sum_{v\in F}(\pi - \theta_{Fv}) = 2\pi$.

N1.2 The boundary need not be geodesic: retain its geodesic-curvature term.

N1.3 Each face is disk-like and all boundary orientations are the induced positive ones.

Export E1/v1. A local curvature identity for each face.

Acceptance gate. Do not silently assume every triangulation edge is geodesic. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N1: Do not silently assume every triangulation edge is geodesic. Recheck dependent nodes N2, N3.

B. N2/v1: Cancel shared edges and collect vertices

Object and role. Pair each edge with its opposite boundary orientation.

Inputs. All local face identities; closed oriented surface; prior exports E1/v1

C — Contract

Own E2/v1: The global integral equals $2\pi V - \pi F$.

K — Knowledge

Retain All local face identities; closed oriented surface; prior exports E1/v1; keep their assumptions and versions attached.

S — Strategy

Pair each edge with its opposite boundary orientation.

V — Validation

No boundary edge may remain unpaired; this uses that M is closed.

E — Execution

Carry out the numbered deductions and publish E2/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N2.1 Sum over the F faces.

N2.2 Every edge belongs to two faces with opposite induced boundary directions; its signed geodesic-curvature integrals cancel.

N2.3 At each vertex the incident interior sectors sum to 2π.

N2.4 There are 3F corner incidences, so the sum of all exterior angles is $3\pi F - 2\pi V$.

N2.5 Thus $\int_M K dA + 3\pi F - 2\pi V = 2\pi F$, yielding $\int_M K dA = 2\pi V - \pi F$.

Export E2/v1. The global integral equals $2\pi V - \pi F$.

Acceptance gate. No boundary edge may remain unpaired; this uses that M is closed. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N2: No boundary edge may remain unpaired; this uses that M is closed. Recheck dependent nodes N3.

C. N3/v1: Convert to Euler characteristic

Object and role. Count incidences before substituting χ.

Inputs. Triangulation counts; prior exports E2/v1

Contract

Own E3/v1: Global Gauss–Bonnet under the stated hypotheses.

Knowledge

Retain Triangulation counts; prior exports E2/v1; keep their assumptions and versions attached.

Strategy

Count incidences before substituting χ.

Validation

A surface with boundary needs additional boundary curvature and corner terms.

Execution

Carry out the numbered deductions and publish E3/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N3.1 Because each triangle has 3 edges and each edge is shared by 2 $triangles, 3F = 2E$.

N3.2 Then $2\pi(V - E + F) = 2\pi V - \pi F$, the preceding integral.

N3.3 Using $\chi(M) = V - E + F$ gives $\int K \, dA = 2\pi\chi(M)$.

N3.4 The formula is independent of the selected compatible triangulation; χ is its topological invariant.

Export E3/v1. Global Gauss–Bonnet under the stated hypotheses.

Acceptance gate. A surface with boundary needs additional boundary curvature and corner terms. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N3: A surface with boundary needs additional boundary curvature and corner terms. Withhold this terminal result until its gate passes.

IV. CONTINUOUS MATHEMATICAL PROOF

Read these steps in order as one proof; the diagram is optional. Numbered activities identify the specific constructions and deductions owned by each node.

A. §4.N1 — Local face identity

1. For each triangular faceF with interior corner angles θ_{Fv}, the local theorem reads $\int_F KdA + \int_\partial F\ k_g ds + \sum_{v\in F}(\pi - \theta_{Fv}) = 2\pi$.
2. The boundary need not be geodesic: retain its geodesic-curvature term.
3. Each face is disk-like and all boundary orientations are the induced positive ones.

B. §4.N2 — Cancel shared edges and collect vertices

1. Sum over the F faces.
2. Every edge belongs to two faces with opposite induced boundary directions; its signed geodesic-curvature integrals cancel.
3. At each vertex the incident interior sectors sum to 2π.
4. There are 3F corner incidences, so the sum of all exterior angles is $3\pi F - 2\pi V$.
5. Thus $\int_M\ KdA + 3\pi F - 2\pi V = 2\pi F$, yielding $\int_M\ KdA = 2\pi V - \pi F$.

C. §4.N3 — Convert to Euler characteristic

1. Because each triangle has 3 edges and each edge is shared by 2 $triangles, 3F = 2E$.
2. Then $2\pi(V - E + F) = 2\pi V - \pi F$, the preceding integral.
3. Using $\chi(M) = V - E + F$ gives $\int K\ dA = 2\pi\chi(M)$.
4. The formula is independent of the selected compatible triangulation; χ is its topological invariant.

□

V. WORKED ORCHESTRATION

For the unit sphere, $K = 1$ and $area4\pi$, so its integral is $4\pi = 2\pi \cdot 2$. A tetrahedral combinatorial triangulation has $V = 4, E = 6, F = 4$ and $\chi = 2$. A flat metric on the torus has $K = 0$ and $\chi = 0$, giving a second consistency check.

A. Boundary and hypothesis audit

A flat unit disk has zero curvature integral but $\chi = 1$; its boundary contributes $\int k_g ds = 2\pi$. Validation rejects applying the closed-surface formula to the disk and restores the boundary terms of the local theorem.

VI. ILLUSTRATIVE FAILURE AND REPAIR

The following is an authored diagnostic scenario. Changed assumptions define a new contract; they do not retroactively prove the original statement.

A. Failure and correction

This is an authored teaching trace, not a recorded agent run.

1. **Contract → Knowledge:** register the §1 theorem, imports, and requested exports before beginning the proof.
2. **Strategy → Execution:** choose the bridges in §3 and carry out the numbered activities in §4; transmit only established E-exports along the §2 DAG.
3. **Validation → responsible node:** test the specific boundary/failure in §5 and the node's individual audit. A failing candidate is withheld; downstream nodes may not use it.
4. **Knowledge + Strategy:** identify whether the error is an omitted condition, an invalid inference, or an attempted extension beyond the theorem. Restore the exact contract or explicitly state a different one.
5. **Execution:** redo the affected mathematical activity using the repaired condition or inference. Recheck consuming nodes, rather than feeding their conclusions back as premises.
6. **Contract:** publish only the supported export with its assumptions and proof location. A hypothetical revised export is v2; the written proof here is the v1 exposition, not a claim that this repair was executed.

B. Hypothesis boundary

A flat unit disk has zero curvature integral but $\chi = 1$; its boundary contributes $\int k_g ds = 2\pi$. Validation rejects applying the closed-surface formula to the disk and restores the boundary terms of the local theorem.

C. Version and recheck

Treat any changed assumption as a new contract. Recheck the revised export and all of its descendants. A backward diagnostic never establishes an earlier theorem. This is an authored teaching trace, not an execution log.

Provenance. This authored proof is complete relative to its declared imports. Reuse each export with its assumptions, proof, and version. The original Markdown remains available for comparison.

Further reading.

- MIT Geometry and Topology: curved geometries and Gauss–Bonnet.